BRAINSPARK

DFM / DFA Analysis

Rear subframe bracket — «hostile» “fixture” × ½ (one chosen route)

CHASSIS · ALUMINIUM A356 (CAST) · DIE CASTING (ALUMINIUM)

GENERATED 2026-08-10 · 3/4 RULES EVALUATED

EXECUTIVE SUMMARY

Die Casting (Aluminium)

Judged against the High-pressure die casting ruleset — 3 of 4 rules could be evaluated on this geometry.

25	4	3	€76,800
MANUFACTURABILITY	FINDINGS	HIGH SEVERITY	PRICED / YEAR

THE PART

No 3D view was captured

The report was generated without an open 3D view. The analysis below is unaffected — it is measured from the solid, not from the picture.

ENVELOPE
248.5 x 162.3 x 88.1 mm

VOLUME
412.7 cm3

SURFACE
1863.2 cm2

FACES
214

VIEW
—

WHAT IS WRONG, WORST FIRST

	FINDING	MEASURED	GUIDELINE	IMPACT / YR
HIGH	Wall thickness outside the die-casting range	30 mm	1–3.5 mm	€76,800
HIGH	Wall area below the minimum die-casting draft	41.2 %	<= 5 %	not priced
HIGH	Undercuts require slides or lifters in the die	3 reg	<= 0 reg	not priced
MEDIUM	Cored hole beyond the core-pin slenderness limit	14.2 L/D	<= 10 L/D	not priced

RULE COVERAGE 75% — 3 of 4 evaluated, 0 could not be measured on this geometry and are NOT passes.

ANALYSIS BASIS

How this report was produced, and from what

READ	MEASURE	JUDGE	PRICE
B-rep from your STEP	kernel + tessellation	rule catalogue	should-cost engine
214 faces	wall p50 30 mm	3 of 4 rules evaluated	€76,800 / yr

MEASURED GEOMETRY

BOUNDING BOX **248.5 x 162.3 x 88.1 mm**	VOLUME **412.7 cm3**
WALL p5 / p50 / p95 **2.1 / 30 / 61.4 mm**	WALL UNIFORMITY **non-uniform**
DRAW DIRECTION **[0, 0, 1]**	UNDERCUT REGIONS **3**
WALL AREA BELOW MIN DRAFT **41.2 %**	UNCLASSIFIED AREA **22.8 %**
HOLES & CUT-OUTS **none**	INTERNAL CUT LENGTH **—**

RECOGNISED RIBS

Thickness is measured at the base, where the 40-60%-of-wall guideline applies; a drafted rib is opened out by the draft term rather than reported at its thinner mid-height. Ratios are against the measured 30 mm nominal wall.

#	THICKNESS	HEIGHT	LENGTH	DRAFT/SIDE	T / WALL	H / WALL
1	4.85 mm	28.4 mm	112.75 mm	1.5 deg	0.16	0.95
2	3.2 mm	19.6 mm	88.4 mm	0 deg	0.11	0.65
3	2.05 mm	14.2 mm	—	2.25 deg	0.07	0.47
4	1.9 mm	33.8 mm	64.1 mm	0 deg	0.06	1.13

ACTIONS

4 decisions, worst first

	FINDING, AND THE CHANGE THAT CLEARS IT	OWNER (ROLE) / DECISION NEEDED
1	**Wall thickness outside the die-casting range** Hold a uniform nominal wall in the 2.0-3.5 mm band and core out heavy sections rather than leaving solid mass. `wall 30 mm -> 3.5 mm`	**Design engineering** Approve the change, or state why the geometry has to stay DUE
2	**Wall area below the minimum die-casting draft** Add at least 1.5 deg of draft to every wall parallel to the draw. `41.2 % of wall area -> <= 5 % of wall area`	**Design + Tooling** Decide without a price — Insufficient draft shortens die life through galling; die life is a tooling-amortisation input, not a geometric one DUE
3	**Undercuts require slides or lifters in the die** Redesign the three occluded faces so they release along the draw, or budget a slide for each. `3 regions -> <= 0 regions`	**Tooling / diemaker** Get a quote: the engines do not price this, only cited literature does DUE
4	**Cored hole beyond the core-pin slenderness limit** Shorten the cored hole to 10 diameters or open it up. `14.2 core L/D -> <= 10 core L/D`	**Design engineering** Decide without a price — Core-pin replacement frequency is a tooling-maintenance input, not a geometric one the engine derives DUE

BY OWNER

Design engineering	2 actions, 1 high severity	EUR 76,800/yr
Design + Tooling	1 action, 1 high severity	not priced
Tooling / diemaker	1 action, 1 high severity	not priced

TOLERANCES

Read from the model

12 dimensions, 3 geometric tolerances and 2 datums were read from the model's semantic PMI. The tightest total band called out is 0.02 mm, and that is what the process-capability rules below are measured against.

```
flatness  0.05 mm
true position  0.1 mm
perpendicularity  0.08 mm
```

HIGH-PRESSURE DIE CASTING

4 findings · 3/4 rules evaluated

MANUFACTURABILITY SCORE	RULE COVERAGE	PRICED IMPACT / YEAR
25 / 100	75%	EUR 76,800

Wall thickness outside the die-casting range

HIGH

MEASURED 30 mm · GUIDELINE 1–3.5 mm

Below about 1 mm the die will not fill reliably before the melt freezes off, and cold shuts appear along the last-to-fill boundary; above about 3.5 mm the section traps porosity as it solidifies from the outside in, holds the cycle open while the centre gives up its heat, and leaves a shrinkage void exactly where the casting is thickest and a machinist is most likely to break into it. Neither failure is recoverable by process tuning once the wall is drawn, which is why wall thickness is the first thing a die caster looks at and the last thing a designer wants to change.

WHAT TO DO

Hold a uniform nominal wall in the 2.0-3.5 mm band and core out heavy sections rather than leaving solid mass. Where local stiffness is needed, add ribs at 40-60% of the nominal wall rather than thickening the wall itself; where a boss must be thick, core it from the back so the section stays even. WARNING_TOKEN_THAT_IS_EXTREMELY_LONG_AND_UNBREAKABLE_ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789

COST IMPACT

wall 30 mm -> 3.5 mm: EUR 12.84 to EUR 12.2 per part, a saving of EUR 0.64 per part (EUR 76,800 per year). Basis: computeShouldCost — cooling-limited cycle (base + k·wall²) and the mass change that comes with it.

SOURCE [INDUSTRY CONSENSUS, no primary source audited]: Aluminium HPDC design guidance (1.0 mm minimum, 2.0–3.5 mm recommended).

Wall area below the minimum die-casting draft

HIGH

MEASURED 41.2 % of wall area · GUIDELINE <= 5 % of wall area

Below about 1 mm the die will not fill reliably before the melt freezes off, and cold shuts appear along the last-to-fill boundary; above about 3.5 mm the section traps porosity as it solidifies from the outside in, holds the cycle open while the centre gives up its heat, and leaves a shrinkage void exactly where the casting is thickest and a machinist is most likely to break into it. Neither failure is recoverable by process tuning once the wall is drawn, which is why wall thickness is the first thing a die caster looks at and the last thing a designer wants to change.

WHAT TO DO

Add at least 1.5 deg of draft to every wall parallel to the draw. Inside walls need roughly twice the outside figure because the casting shrinks onto them.

COST IMPACT

Not priced. Insufficient draft shortens die life through galling; die life is a tooling-amortisation input, not a geometric one the engine derives.

SOURCE [INDUSTRY CONSENSUS, no primary source audited]: Aluminium HPDC design guidance (1.0 mm minimum, 2.0–3.5 mm recommended).

Cored hole beyond the core-pin slenderness limit

MEDIUM

MEASURED 14.2 core L/D · GUIDELINE <= 10 core L/D

Below about 1 mm the die will not fill reliably before the melt freezes off, and cold shuts appear along the last-to-fill boundary; above about 3.5 mm the section traps porosity as it solidifies from the outside in, holds the cycle open while the centre gives up its heat, and leaves a shrinkage void exactly where the casting is thickest and a machinist is most likely to break into it. Neither failure is recoverable by process tuning once the wall is drawn, which is why wall thickness is the first thing a die caster looks at and the last thing a designer wants to change.

WHAT TO DO

Shorten the cored hole to 10 diameters or open it up. A drilled secondary operation is the alternative and it carries its own cost.

COST IMPACT

Not priced. Core-pin replacement frequency is a tooling-maintenance input, not a geometric one the engine derives.

3 of 9 checked features break this:

- 2x Ø5 x71 14.2 core L/D at (43, -18, 6)
- Ø6.5 x83.2 12.8 core L/D at (-105, 63, 33)
- 4x Ø10 x104 10.4 core L/D at (0, 0, -12)

SOURCE [YOUR COMPANY STANDARD, set in this workspace]: Company standard: our tool room will not run a core pin...
THRESHOLD: resolved for company standard

Undercuts require slides or lifters in the die

HIGH

MEASURED 3 regions · GUIDELINE <= 0 regions

Below about 1 mm the die will not fill reliably before the melt freezes off, and cold shuts appear along the last-to-fill boundary; above about 3.5 mm the section traps porosity as it solidifies from the outside in, holds the cycle open while the centre gives up its heat, and leaves a shrinkage void exactly where the casting is thickest and a machinist is most likely to break into it. Neither failure is recoverable by process tuning once the wall is drawn, which is why wall thickness is the first thing a die caster looks at and the last thing a designer wants to change.

WHAT TO DO

Redesign the three occluded faces so they release along the draw, or budget a slide for each. A lifter is cheaper than a slide where the feature is shallow.

COST IMPACT

Not priced. Slide and loose-core tooling is quoted by the diemaker; the piece-price engines do not model it.

Industry guidance puts a side action, lifter or collapsible core at roughly $500–$5,000 of tooling per feature. Cited literature, NOT an engine result — confirm with your toolmaker.

SOURCE [INDUSTRY CONSENSUS, no primary source audited]: Aluminium HPDC design guidance (1.0 mm minimum, 2.0–3.5 mm recommended).

PASSED (1)

- Sharp internal corners concentrate stress and restrict flow — measured 2.4 mm against >= 1.6 mm

ALTERNATIVE ROUTES

Your route is Die Casting (Aluminium), against the other 4

The findings above are YOUR route and nobody else's. This page exists only to answer the next question — would another process make the same geometry for less. Each row is the same measured geometry run through THAT process's own rule family, priced by computeShouldCost and carbon-scored by computeCarbon on the cost engine's own input mass. Nothing is blended into a single ranking.

Process	Score	Rules	EUR/part	Tooling	kg CO2e
Extrusion	50	3/3	EUR 3.18	EUR 25,000	5.20
-EUR 3.42/part vs your route, -EUR 116,458 tooling · 2 high-severity					
Die Casting (Aluminium)	**13**	**5/9**	**EUR 6.60**	**EUR 141,458**	**7.48**
YOUR ROUTE — the findings above are this one · 3 high-severity · Only 5 of 9 rules could be evaluated, so this score rest...					
Sand Casting	33	10/10	EUR 7.55	EUR 28,292	8.10
+EUR 0.95/part vs your route, -EUR 113,167 tooling · 2 high-severity					
Machining (CNC)	100	3/5	EUR 27.54	EUR 4,000	9.98
+EUR 20.94/part vs your route, -EUR 137,458 tooling					
Hydroforming	—	0/3	not priced	—	—
No rule in this family could be evaluated on this geometry, so there is no score — not a clean sheet.					

Extrusion prices 3.42 EUR/part below Die Casting (Aluminium) on lower tooling (EUR 25,000 against EUR 141,458), and scores 50 over 3 of 3 rules with 2 high-severity findings. That is a quotation to ask for, not a decision this tool has taken for you.

Not applicable to undefined: Injection Moulding, Die Casting (Zinc).

DESIGN FOR ASSEMBLY

6 parts · 3 distinct types

6	41.8 s	withheld	withheld
PARTS	ASSEMBLY TIME	THEORETICAL MIN	DFA INDEX

5 of 5 parts have unanswered DFA questions, so the theoretical minimum and design efficiency are withheld. Handling and insertion times below are geometric and stand on their own.

Handling and insertion times come from the brainspark-dfa-v1 model. MTM-structured; BrainSpark coefficients; calibratable. NOT Boothroyd & Dewhurst tables.

#	PART	OFF	a+b	HANDLE s	INSERT s	TOTAL s
0	bracket casting	1	720	3.24	1.5	4.74
1	reinforcement plate — very long...	1	360	1.63	1.5	3.13
2	M10 bolt	4	180	1.13	6.5	7.63
3	washer	4	180	1.83	1.5	3.33
4	bush	2	180	1.13	2.5	3.63

Suspected fasteners (geometric signature only, confirm against the BOM): solid-3 (medium), solid-4 (medium).

APPENDIX

Where every number came from

GEOMETRY

Measured by an OpenCascade kernel. Draft angles, undercut classification and wall thickness on the tessellation — which works on freeform surfaces, where a plane-and-cylinder analysis measures almost nothing on a real cast or moulded part. Holes, apertures and features from the B-rep topology. The AI wrote none of the numbers in this report.

DRAW DIRECTION

Chosen by sweeping candidate axes and scoring undercut area, not assumed. Undercuts are separated from zero-draft drag faces: a zero-draft wall is fixable with a degree of taper, an undercut buys a slide or a lifter, and reporting them as one number would overstate the tooling problem.

COST

Re-run through the same deterministic engines used elsewhere in BrainSpark, once with the geometry as drawn and once with the rule satisfied. Findings the engines cannot price say so, with the reason.

THRESHOLDS

Every DIMENSION here was measured from your file and is reproducible. The GUIDELINES they are compared against are not of the same standing. The catalogue-wide counts were not sent with this analysis, so they are not stated here — each finding still carries its own grade beside its source, which is where the claim that matters for that finding lives. Every finding carries its grade. Treat a finding as a screening result that opens a conversation with your supplier — not as a specification.

THREE OUTCOMES, NOT TWO

Failed, passed, and NOT EVALUATED. A rule whose measurement this part does not provide is listed as unevaluated with the reason, never as a pass — so the coverage figure beside the score tells you how much of the catalogue actually ran.

DFA

Handling times follow the published Boothroyd-Dewhurst METHOD. Their tables are copyrighted and are not reproduced; the coefficients here are ours and should be calibrated against your own line data before a labour commitment. Part symmetry is MEASURED by rotating each solid and intersecting it with itself, not inferred from inertia. The three DFA questions concern function and intent, which a static solid cannot answer — until every part is answered, the theoretical minimum and the index are withheld.

WHAT THIS DOES NOT COVER

Threads are not recognised. GD&T is read only where the file carries semantic AP242 PMI. Ribs are found from opposed planar side faces on a planar base, so a rib with curved sides is not checked. Surface finish and material come from your input, not from the model. The tool-reach sweep does not model the holder, the spindle nose or the machine envelope, so a face it calls reachable may still not be.